

2nd Part Problem Set for Applied Abstract Algebra
Auslander-Reiten quivers

1. Let $M = (M_i, \varphi_\alpha) \in \text{rep } Q$ and $d_i = \dim M_i$. Show that

$$0 \rightarrow M \rightarrow \bigoplus_{i \in Q_0} d_i I(i) \rightarrow \bigoplus_{\alpha: i \rightarrow j \in Q_1} d_j I(i) \rightarrow 0$$

is an injective resolution for M .

2. Let $M \in \text{rep } Q$ and

$$0 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

be standard projective resolution. Show that

$$0 \rightarrow D(M) \rightarrow D(P_0) \rightarrow D(P_1) \rightarrow 0$$

is the injective resolution for $D(M) \in \text{rep } Q^{op}$ defined in exercise 1. Show that duality functor D takes a minimal projective resolution for M to a injective envelope for $D(M)$.

3. (Check exercise 6 of first exercise sheet.) Let Q be the quiver

$$1 \begin{array}{c} \xleftarrow{\alpha} \\ \xleftarrow{\beta} \end{array} 2$$

For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$1 \begin{array}{c} \xleftarrow{1} \\ \xleftarrow{\lambda} \end{array} 2 \quad \lambda \in k;$$

$$1 \begin{array}{c} \xleftarrow{0} \\ \xleftarrow{1} \end{array} 2 \quad \lambda = \infty.$$

Recall that

$$S(1) : k \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} 0 \quad S(2) : 0 \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} k.$$

- (a) Show that the short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & S(1) & \longrightarrow & M_\lambda & \longrightarrow & S(2) \longrightarrow 0, \\ 0 & \longrightarrow & S(1) & \longrightarrow & M_\mu & \longrightarrow & S(2) \longrightarrow 0, \end{array}$$

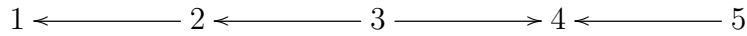
are not equivalent if $\lambda \neq \mu$.

- (b) Show that $\tau M_\lambda = M_\lambda$ for all λ .

4. Prove that P is projective if and only if $\text{Ext}^1(P, N) = 0$ for all representations N .

5. Show that isoclasses of indecomposable representations of quivers of type A_n are determined by their dimension vectors of the form $(0, \dots, 0, 1, \dots, 1, 0, \dots, 0)$, and the corresponding representation is $M = (M_i, \varphi_\alpha)$ with $M_i = k$ if the dimension at i is one, and $M_i = 0$ otherwise; and $\varphi_\alpha = 1$ if the dimension at $s(\alpha)$ and at $t(\alpha)$ is one, and $\varphi_\alpha = 0$ otherwise.

6. Compute the Auslander-Reiten quiver for the following quiver:



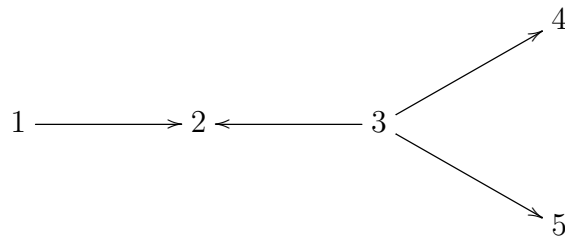
7. Compute the Auslander-Reiten quiver for the following quiver:



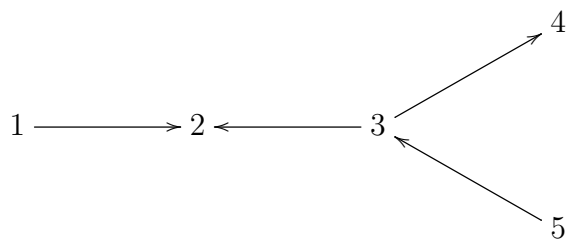
8. The Cartan matrix $C = \{c_{ij}\}_{1 \leq i, j \leq n}$ of the quiver Q , where c_{ij} is the number of paths from j to i and n is the number of vertices in Q . Show that C is invertible and find the inverse matrix C^{-1} (See section 3.1.2.2 of the book).

9. Let Q be a quiver of type A with n vertices and consider triangulation of corresponding to Q . Show that if we consider an elementary clockwise rotation of the diagonal corresponding to 1 we obtain the diagonal corresponding to $P(1)$. Show that if we consider an elementary counterclockwise rotation of the diagonal corresponding to 1 we obtain the diagonal corresponding to $I(1)$.

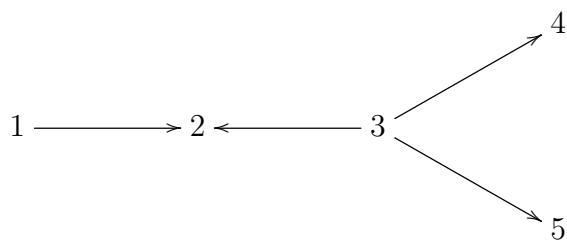
10. Compute the Auslander-Reiten quiver for the following quiver:



11. Compute the Auslander-Reiten quiver for the following quiver:



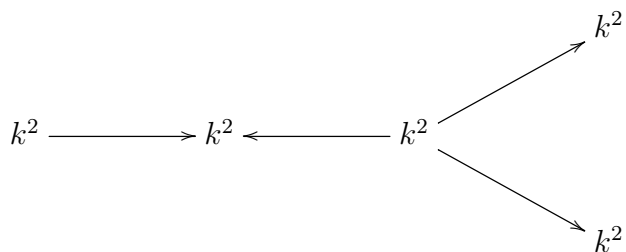
12. Let Q be the quiver



and consider the indecomposable representations $\dim L = (0, 1, 1, 1, 0)$ and $N = (1, 1, 2, 1, 1)$. Prove that there is a unique representation M for which there exists a non-split short exact sequence

$$0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$$

13. Decompose the following representation of Q into direct sum of indecomposable representations



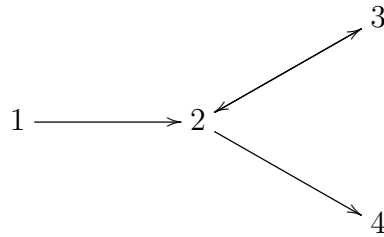
all maps are identities.

14. Let Q be a quiver of type D with n vertices and consider triangulation of corresponding to Q . Show that if we consider an elementary clockwise rotation plus tagging of the diagonals corresponding to $n - 1$ and n we obtain the diagonals corresponding to $P(n - 1)$ and $P(n)$ respectively.

15. Construct the decomposition of $E_{\mathbf{d}}$ into orbits for

$$Q : \quad 1 \longrightarrow 2 \longleftarrow 3 \quad \mathbf{d} = (2, 3, 1).$$

16. Construct the decomposition of $E_{\mathbf{d}}$ into orbits for $\mathbf{d} = (2, 2, 1, 1)$ and Q



17. Let $q : \mathbb{Z}^n \rightarrow \mathbb{Z}$ be a quadratic form. Show that the form

$$(\mathbf{x}, \mathbf{y}) = q(\mathbf{x} + \mathbf{y}) - q(\mathbf{x}) - q(\mathbf{y})$$

is symmetric and bilinear.

For the following three exercises we need to define a *tree*, *wild quiver* and *valence* of a vertex in a quiver. A quiver Q is called a *tree* if its underlying graph has no cycles. Note that parallel edges are cycles. A quiver without oriented cycles which is neither Dynkin nor Euclidean is called *wild*. The *valence* of a vertex in a quiver or graph is the number of its neighbor vertices.

18. Prove that for $n \leq 5$ any tree quiver is Dynkin or Euclidean.
19. Show that the smallest wild tree with valence at most 3 at each vertex has 7 vertices.
20. Show that the smallest wild tree that has at most one vertex of valence 3 at each vertex has 8 vertices.